Sven Weber

Postdoctoral Researcher at UC Berkeley - EVOLAB

Oakland, CA, United States

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Experience and Education

Postdoctoral Fellow

Berkeley, CA

EVOLAB — UC Berkeley

Jan 2025 - Dec 2026

- Walter Benjamin Fellowship on long-term biodiversity change
  - Designed & led decade-scale comparisons across the Hawaiian archipelago
  - Studied plant–arthropod interaction networks
  - Investigated signals of tripartite networks
  - Pilot studies to investigate pathogenic and invasive nematodes on Hawai'i
  - Funded by the DFG Postdoctoral Fellowship

Postdoctoral Researcher

Berkeley, CA

EVOLAB; Essig Museum of Entomology — UC Berkeley

Jul 2023 - Dec 2024

- PROJECT: EDIAs (Early Detection of Invasive Arthropods)
  - Developed eDNA-based methods for biodiversity monitoring
  - Led fieldwork in Hawai'i
  - Investigating of adaptive radiations of spiders on the Pacific islands
  - Reference data base compilation for Hawiian taxa

Researcher

Trier, GER

Department of Environmental Sciences — Trier University

Jan 2023 - May 2023

- PROJECT: Collaboration with Bayer AG on eDNA-based pollinator monitoring
  - Designed workflows to detect changes in pollinator communities
  - Coordinated KICK-OFF for academic–industry cooperation

Dr. rer. nat.

Trier, GER

Krehenwinkel Lab — Trier University

Apr 2020 - Dec 2022

- THESIS: Utility of metabarcoding natural eDNA samplers for retrospective biodiversity monitoring
  - Developed assays for gut content (spiders & mussels)
  - Developed eDNA-based approaches for leaf samples, and museum-derived samples like herbaria
  - Gained in-depth expertise in bioinformatics and biodiversity trend analysis of time series
  - Funded by DBU doctoral scholarship

MSc Environmental Sciences

Aachen, GER

Ross-Nickoll Lab — RWTH Aachen

Oct 2018 - Sep 2019

- THESIS: Spider Biodiversity changes across fragmented agricultural landscapes
  - Conducted temporal analyses of long-term observational data
  - Trained in community ecology, field surveys, and multivariate statistics
  - In-depth knowledge of central european spider fauna
  - Supported by International Society of Arachnology student grant

MSc Molecular Ecology

Düsseldorf, GER

PopGen Lab — Heinrich Heine University

Oct 2017 - Sep 2019

- THESIS: Spider Biodiversity changes across fragmented agricultural landscapes
  - Trained in Sensory Ecology
  - CRISPR/Cas9 gene editing in *Arabidopsis thaliana*
  - Learning in-depth about miRNA expression in *Solanum lycopersicum*
  - Trained in plant immune responses in plant-pathogen interactions

BSc Biology

Düsseldorf, GER

PopGen Lab — Heinrich Heine University

Oct 2010 - Sep 2017

- THESIS: Fitness differences of *Arabidopsis thaliana* on heavy metal soils
  - Built foundation in plant genetics, lab protocols, and data analysis (PERL & R)
  - Investigated population structure in *Solanum lycopersicum*
  - Developed early skills in molecular ecology and evolution

Publications in Peer-Reviewed Journals

10. Pfau, M., **Weber, S.**, Kennedy, S. […], Gillespie, R. **(2025)**. *Do invasive spiders escape their native microbiomes? Patterns of microbial variation in native and invasive spiders in Hawai'i*. Ecology and Evolution 15 (10), e72175 .

9. Banerjee, P. […], **Weber, S.**, Chen, C. (2025). *Revealing the potential for environmental DNA (eDNA) to detect species interactions in two economically important fruits: Carica papaya and Ananas comosus*. Environmental DNA.

8. Junk, I. & Hans, J., **Weber, S.**, Melcher, A. […], Krehenwinkel, H. **(2025)**. *Archived natural DNA samplers reveal four decades of biodiversity change across the tree of life*. Nature Ecology & Evolution .

7. Melcher, A., **Weber, S.**, Birkhofer, K., Harms, D., […], Krehenwinkel, H., **(2024)**. *To pool or not to pool - Pooled metabarcoding does not affect estimates of prey diversity in spider gut content analysis*. Ecological Entomology .

6. Stothut, M., Mahla, L., Backes, L., **Weber, S.**, […], Krehenwinkel, H. **(2024)**. *Recovering plant-associated arthropod communities by eDNA metabarcoding historical herbarium specimens*. Current Biology .

5. Gajski, D., Wolff, J., Melcher, A., **Weber, S.**, […], Krehenwinkel, H. **(2024)**. *Facilitating taxonomy and phylogenetics: An informative and cost-effective protocol integrating long amplicon PCRs and third-generation sequencing*. Molecular Phylogenetics and Evolution 192, 107988.

4. **Weber, S.** & Stothut, M., Mahla, L., […], Krehenwinkel, H. (2024). *Plant-derived environmental DNA complements diversity estimates from traditional arthropod monitoring methods but outperforms them detecting plant–arthropod interactions*. Molecular Ecology Resources 24 (2), e13900.

3. **Weber, S.**, Junk, I., Brink, L., Woerner, M., […], Krehenwinkel, H. **(2023)**. *Molecular diet analysis in mussels and other metazoan filter feeders and an assessment of their utility as natural eDNA samplers*. Molecular Ecology and Resources 23 (2), 471-485.

2. Krehenwinkel, H & **Weber, S.**, Broekmann, R., Melcher, A., Hans, J. […], Veith, M. **(2023)**. *Environmental DNA from archived leaves reveals widespread temporal turnover and biotic homogenization in forest arthropod communities*. Elife 11, e78521.

1. Krehenwinkel, H., **Weber, S.**, Künzel, S., Kennedy, S. **(2022)**. *The bug in a teacup—monitoring arthropod–plant associations with environmental DNA from dried plant material*. Biology Letters 18 (6), 20220091 .

Publications submitted or under Peer-Review

6. **Weber, S.**, & Scali, E., […], Gillespie, R., Roderick, G. **(2025)**. *Shared Landscapes, Divergent Communities: Fungi and Arthropods Follow Different Assembly Rules*. Ecography .

5. **Weber, S.** & Hutchins, L. […], Roderick, H. **(2025)**. *Topography structures of arthropod communities revealed by leaf-derived environmental DNA on O'ahu, Hawai'i*](`https://www.biorxiv.org/content/10.64898/2026.04.09.716978v1`). Environmental DNA.

4. Banerjee, P. & **Weber, S.**, […], Gillespie, R., Roderick, G. **(2025)**. *Littered with information - eDNA biodiversity studies enabled by leaf litter samples from Hawaiian montane wet forests*. Molecular Ecology and Resources DNA.

3. Banerjee, P., […], **Weber, S.**, Gillespie, R., Roderick, G. **(2025)**. *Comparison of environmental DNA and bulk DNA metabarcoding for assessing terrestrial arthropod diversity across three habitat types on Guam*. Molecular Ecology and Resources.

2. Boren, A. M. & **Weber, S.**, […], Gillespie, R., Roderick, G., Roy, K. **(2025)**. *Varying and opportunistic symbioses between ambrosia beetles and fungal pathogens can facilitate recent emergence of Rapid 'Ōhi'a Death in Hawai'i*. Conservation Biology .

1. **Weber, S.** & Melcher, A., Calaor, J. Kennedy, S. […], Krehenwinkel, H. **(2024)**. *Spiders as natural DNA samplers can recover arthropod community diversity and biotic interactions across space and time*. Molecular Ecology and Resources.

Invited Talks

Weber,S. (05/2026). *Does Ecological Speciation Drive Diversification? An overview about Hawaiian* Lycosidae *spiders*. Talk scheduled for the Lycosids of the Wold Workshop at the Southwest University, Chongqing - China.

Weber,S. (04/2026). *eDNA-based monitoring to answer ecological questions*. Talk scheduled for the Ecolunch at the University of California, Berkeley - USA.

Weber, S. (10/2023). *Terrestrial environmental DNA – A chance for retrospective biomonitoring by using biological archives*. Talk presented at Essig Museum of Entomology of the University of California, Berkeley - USA.

Weber, S., Junk, I., Hans, J., Stothut, M., Mahla, L., Klein, R., Paulus, M., Veith, M., Krehenwinkel, H. (03/2023) *Environmental specimen banks and other biological archives for retrospective biodiversity studies*. Talk presented at LTER-D Yearly meeting - online.

Weber, S. (01/2023). *eDNA-based retrospective monitoring of canopy associated arthropod communities*. Talk presented at Meeting of Environmental Specimen Banks - Trier, Germany.

Weber, S. (11/2022) *Der Käfer in der Teetasse*. Talk presented at the 25th Autumn Meeting at the Natural History Museum St. Gallen - Switzerland.

Weber, S. (04/2022) *Retrospective monitoring of arthropod communities and more*. Talk presented at the Senckenberg Natural History Museum Seminar, online.

Weber, S. & Melcher, A. (12/2021). Biodiversity assessments in the 21st century and what spiders have to do with it. **Plenary talk** given at the Asian Society Arachnology, ASA 6th meeting – online (COVID)

Weber, S. & Melcher, A. (09/2021). *DNA-gestützte Analysen von Nahrung heimischer Spinnen*. **Plenary talk** presented at Arachnoweekend of the DeArGe meeting, online (COVID).

Weber, S. (08/2021). *Retrospective monitoring of plant-associated arthropod communities*. Talk presented at the German Federal Agency of Nature Conservation: Convention on Biological Diversity - Vilm, Germany.

Weber, S. (05/2021). *Entwicklung eines Umwelt-DNA (eDNA) basierten Monitorings heimischer, pflanzenassoziierter Insektengemeinschaften* Talk presented for

the German Federal Environmental Foundation yearly meeting (DBU) – online (COVID).

Conference Talks

Weber, S. & SERDP Pacific EDIAs Consortium (12/2025) *Mycangia Matter: Ambrosia Beetles and ROD*. Talk presented at PEBM - Honolulu USA.

Weber, S. & SERDP Pacific EDIAs Consortium (9/2024). *Littered with information - eDNA biodiversity studies enabled by leaf litter samples from Hawaiian montane wet forests*. Talk presented at the iBOL conference - Belem, Brasil.

Weber, S., Melcher, A., Stothut, M., Mahla, L., Hirschler, L., Kripp, A., Krehenwinkel, H. (09/2022). *eDNA – a new application for monitoring spider diversity?* Talk presented at the 33rd European Congress of Arachnology (ECA) – Greifswald, Germany.

Weber, S. (11/2022). *Der Käfer in der Teetasse*. Talk presented at the 34th Westdeutscher Entomologentag - Düsseldorf, Germany.

Weber, S. (11/2021). *eDNA basiertes monitoring von Arthropoden*. Talk presented at the 33rd Westdeutscher Entomologentag - online (COVID).

Weber, S., Melcher, A., Künzel, S., Broekmann, R., Paulus, M., Klein, R., Krehenwinkel, H. (08/2021). *eDNA-based monitoring of canopy associated arthropod communities* European Congress of Arachnology (ECA): European Congress of Arachnology 32nd Meeting – online (COVID).

Weber, S. (06/2020). *Long term monitoring of canopy associated spider communities by eDNA metabarcoding from leaves*. American Arachnological Society (AAS) - online (COVID).

Recognitions And Awards

2025 - 2026	DFG - German Research Foundation - Walter Benjamin Postdoctoral Fellowship	*$135,000*
2021 - 2022	DBU - German Federal Environment Agency - Doctoral Scholarship	*$43,500*
2018	ISA - International Society of Arachnology - Oscar and Jan Francke Student Research Fund	*$870*

Teaching Experience

Main Instructor

Trier University

- Programming and metabarcoding analyses in R
- Biomonitoring
- Information Gathering & Scientific Presentation

Teaching Assistant

Trier University

- Anatomy and Diversity of the Animal Kingdom

Teaching Assistant

Heinrich-Heine-Universität, Düsseldorf

- Zoologie
- Molekulare Biologie
- Biologie für Mediziner

Academic Services

Supervision of over 15 Students since 2020:

Degree	Student Name - My Role - Year	Current Position
Honor's Thesis		
UC Berkeley	Nimrah Khan - Main Supervisor - 2026	Undergraduate Student UC Berkeley, USA
UC Berkeley	Rachel Chen - Main Supervisor - 2025	Undergraduate Student UC Berkeley, USA
UC Berkeley	Maia Supple - Main Supervisor - 2025	PhD Student UCLA, USA
UC Berkeley	Alex M. Boren - Main Supervisor - 2024	Research Assistant UC Berkeley, USA
UC Berkeley	Alex A. Farrow - Main Supervisor - 2023	Research Assistant UCSF, USA
Master's Thesis		
UC Berkeley	Tim Hörmann - Co Supervised - 2025	Master's Konstanz University, Germany
Trier University	Arndt Schmidt - Co Supervised - 2023	PhD Student Trier University, Germany
Trier University	Lena Hirschler - Main Supervisor - 2022	Research Assistant GAIAC, Germany
Trier University	Manuel Stothut - Main Supervisor - 2022	Researcher ESB Trier, Germany
Trier University	Lisa Mahla - Main Supervisor - 2022	PhD Student Trier University, Germany
Trier University	Alanah Kripp - Main Supervisor - 2022	PhD Student RPTU, Germany
Trier University	Lukas Brink - Main Supervisor - 2021	Left academia
Bachelor's Thesis		
Trier University	Tim Hörmann - Co Supervised - 2022	Master's Konstanz University, Germany
Trier University	Lilly Menzl - Co Supervised - 2021	Master's Freiburg University, Germany
Trier University	Ferdinand Meiß - Co Supervised - 2021	Left academia
Trier University	Adrienne Opala - Co Supervised - 2020	Left academia
Trier University	Dana Drück - Co Supervised - 2020	Left academia
Trier University	Gianis Raubuch - Main Supervisor - 2020	Left academia

Peer Reviews

25 reviews in 15 different journals since 2021 - 11/2025 (alphabetical order):

Arachnologische Mitteilungen, Biodiversity and Conservation, Ecological Genetics and Genomics, Ecography(2), environmental DNA (3), Frontiers of Zoology (2), Journal of Arachnology, Journal of Insect Conservation (2), Molecular Ecology and Resources (4), Scientific Reports, peerJ, Thalassas: An International Journal of Marine Sciences, The Journal of Basic and Applied Zoology, Zoosystematics and Evolution (2).

References

Prof. Dr. Rosemary Gillespie — *USA*
Postdoctoral Research Advisor — *University of California Berkeley*

Prof. Dr. George Roderick — *USA*
Postdoctoral Research Advisor — *University of California Berkeley*

Prof. Dr. Henrik Krehenwinkel — *GER*
Dissertation Advisor — *Trier University*

Prof. Dr. Klaus Birkhofer — *GER*
Collaborator — *B-TU*

Prof. Dr. Martina Roß-Nickoll — *GER*
Master's Advisor — *RWTH Aachen*

Prof. Dr. Laura E. Rose — *GER*
Undergraduate Advisor — *Heinrich-Heine University*